

GAS-OPERATED MACHINE-GUNS

Gas-operated machine-guns harness the energy of the gases produced by the exploding cartridge (see p.305) that propel the bullet down the barrel. A portion of these gases is bled off to reload the gun by driving a piston to the rear, which pushes the bolt backward. This extracts the spent cartridge and chambers a new one. Guns using this system can be made light and easily to control because the gas piston and springs inside the gun absorb much of the recoil. These machine-guns evolved in the 1880s and 1890s, and the first claim to a working design was the Colt-Browning “Potato Digger.” A more sophisticated design belonged to the Hotchkiss company’s hugely successful Hotchkiss machine-gun. Gas-operated systems have continued to proliferate.

▼ LEWIS GUN M1914

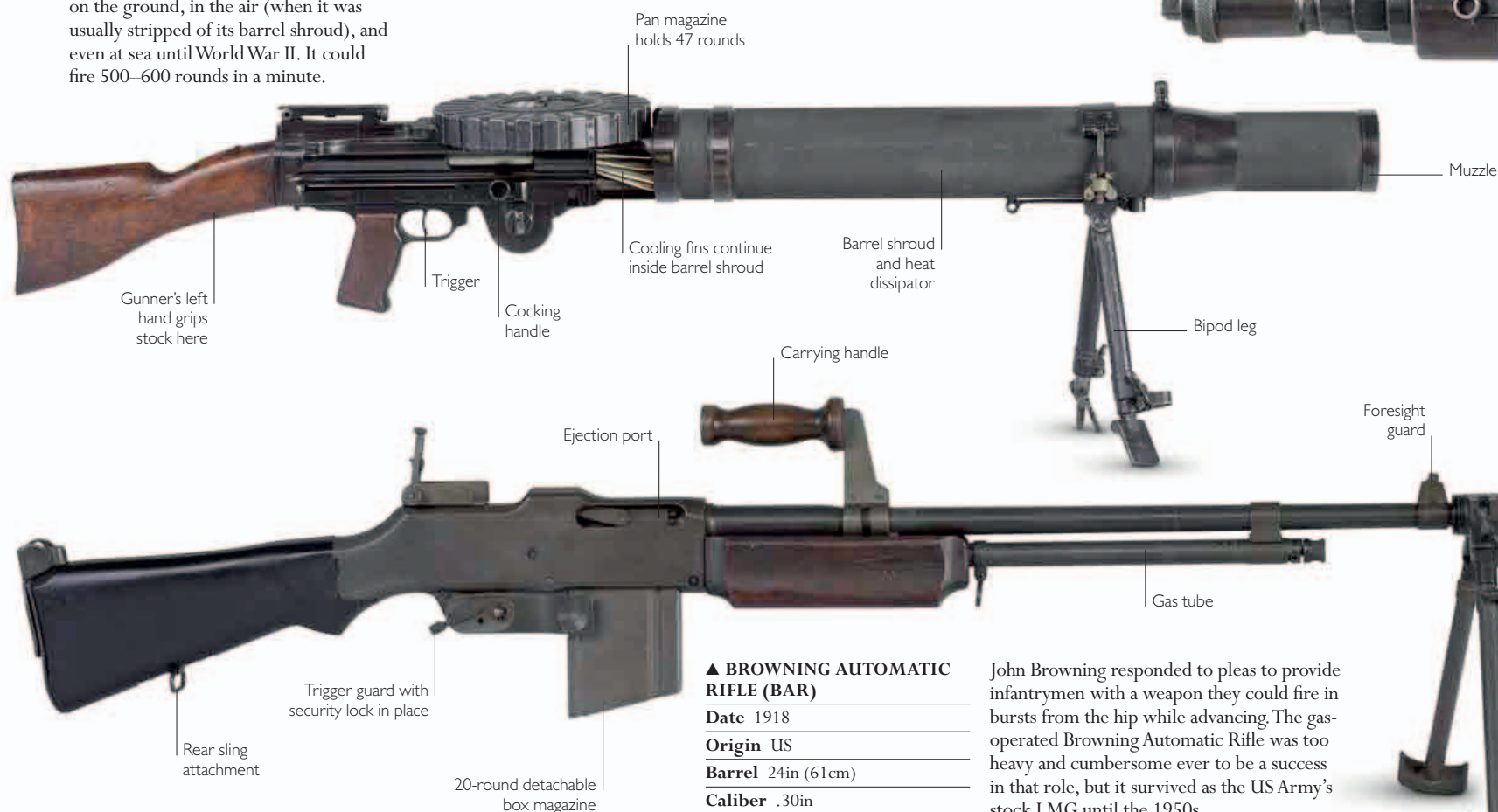
Date 1914

Origin US

Barrel 26in (66cm)

Caliber .303in

The air-cooled Lewis gun was the first light machine-gun (LMG) used on the Western Front. Taken up by the Belgians, then by the British, it remained in service on the ground, in the air (when it was usually stripped of its barrel shroud), and even at sea until World War II. It could fire 500–600 rounds in a minute.



▲ COLT-BROWNING M1895 “POTATO DIGGER”

Date 1895

Origin US

Barrel 28½in (72cm)

Caliber .30in Krag



The Colt M1895 was the creation of John Browning; it was nicknamed the “Potato Digger” because of its innovative mechanics. Some of the gas produced by the exploding charge was tapped off from near the muzzle to drive an arm through a 170-degree action. Through a linkage, the arm, in turn, powered the opening and closing of the breech. The M1895 was reliable enough, and served with the US Army, Navy, and Marine Corps around the turn of the century.

▲ BROWNING AUTOMATIC RIFLE (BAR)

Date 1918

Origin US

Barrel 24in (61cm)

Caliber .30in

John Browning responded to pleas to provide infantrymen with a weapon they could fire in bursts from the hip while advancing. The gas-operated Browning Automatic Rifle was too heavy and cumbersome ever to be a success in that role, but it survived as the US Army’s stock LMG until the 1950s.